

An Exploratory Analysis of Turkish TV Series Ratings and Prediction Using Machine Learning Models

DSA210 Term Project Final Report

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1. Introduction & Motivation

Turkish TV Series, much like other forms of television programming around the world, provide significant amounts of content to the global entertainment market. A variety of viewership trends exist today globally. As more and more people access entertainment content through traditional television viewing options and increasingly popular streaming services, international interest in Turkish TV Series is expanding rapidly. This growing trend creates a need to study the various factors affecting Turkish TV Series ratings.

A primary reason behind the focus of this project was to investigate the potential relationships between some key features of Turkish TV Series (such as the amount of user-generated vote totals received by each series, the specific genre or genres of each series, each series' length (runtime), and the year each series premiered) and their respective ratings listed on the IMDb. If we find statistically meaningful relationships between the variables I mentioned above and ratings, then we will be able to better understand how TV Series ratings are influenced, including identifying those variables most strongly associated with high or low ratings. Another purpose of this project was to practice applying Data Science methodologies to real-world datasets. The project included using exploratory data analysis, statistical hypothesis testing, and predictive modeling to analyze the data from multiple viewpoints.

Additionally, this project compared performance of two types of Machine Learning Models. Specifically, Linear Regression Models and Random Forest Models were used and analyzed to determine how well they could be utilized to make predictions about TV Series ratings based upon the provided data features.

2. Data Source and Data Collection

2.1 Dataset Source

The public dataset that has been used in the project has been provided through Kaggle with respect to various aspects related to IMDb and Turkish television shows. This data includes; the genre, vote count, year of release, run time and IMDb rating for each show.

The data mentioned above is chosen for two reasons. First, the data includes variables which can be classified into both numeric and categorical types which are suitable for statistical analysis and machine learning applications. Second, IMDb is one of the most popular ways people use to determine the quality of a TV show. Therefore, the data set was an ideal choice to study how people rate TV series.

2.2 Data Cleaning and Preprocessing

Before analysis the dataset was preprocessed to make it suitable for statistical analysis and machine learning.

First, unnecessary or incomplete observations were removed from the dataset. Important variables like ratings, votes, runtime, start year, and genres had to be filled in so rows with insufficient information were removed from the dataset. Next, votes, runtimeMinutes, startYear, and the numerical variable rating were converted to numerical data types using pandas functions, as some values were stored as text values. In addition, genre information also had to be preprocessed, since multiple genres could belong to a single TV series. The genre values were turned into machine-readable binary variables using encoding for inclusion as categorical values in the machine learning models. After preprocessing, the cleaned dataset was used for exploratory data analysis, hypothesis testing, and predictive modeling.

3. Exploratory Data Analysis (EDA)

3.1 Descriptive Statistics

Exploratory Data Analysis (EDA) was done for a better understanding of the datasets structure and characteristics before the application of statistical test and machine learning model. The descriptive statistic was applied for describing the most important features of the variables that were analyzed.

This dataset contains multiple types of information about Turkish TV series. Both numerical and categorical variables exist in this dataset. Numerical variables, such as ratings, votes, run time, and release years were analyzed using the statistics mean, median, minimum, maximum and standart deviation.

Descriptive results indicated that most IMDB ratings fall within a moderately distributed area; however, vote totals vary greatly from one series to another. These findings suggest that a subset of series generates substantially greater audience engagement then other series.

3.2 Distribution of Ratings

A histogram was used with a visual comparison to determine how ratings for most TV shows are distributed across different levels. In general, the data indicates that many of the ratings for the majority of TV shows fall into the medium-to-high value ranges; therefore, there is a lesser amount of rating evidence that would indicate extreme lower ratings for TV shows in this data set.

In addition, the data suggests an imperfect symmetry in the distribution of ratings for audience input as well as some degree of skewness toward higher-rated values. It can be assumed from the above data that popular or successful TV shows tend to have higher ratings on IMDb.

3.3 Relationship Between Votes and Ratings

The relationship between vote count and ratings for TV show on IMDB was assessed through an examination of correlations along with examining scatter plots. The correlation results showed that there is a weak positive relationship between the number of votes and the ratings of TV shows. Therefore, it appears that those TV shows receiving more engagement from audiences are generally rated slightly better than other TV shows.

However; since there was found to be such a small correlation, it cannot be concluded that the total amount of voting will accurately predict the ratings of a TV show alone so it can be told that it is influenced by some additional variables.

3.4 Genre Analysis

Genre Analysis was completed to determine how the most popular TVshow genres relate to the overall average ratings. The top three most common genres in this data set are Drama, Comedy and Romance.

The average ratings by genre revealed a relatively small variation among the genres. Although descriptive analysis did not reveal an extreme difference in ratings between genres, genre information is thought to be a potential valuable information for prediction after proper preprocessing and encoding.

3.5 Runtime and Start Year Analysis

Runtime length as well as the release year for each of these series was looked at through scatter plots and correlation analysis to see if there was any relationship with ratings. These methods indicated a weak relationship between runtime and ratings. The data from this analysis also showed no significant association with ratings based on the start year of production; therefore, it appears that newer TV series do not necessarily receive significantly higher or lower ratings compared to older series. Overall, this EDA process assisted greatly in gaining insight into the data, and in identifying the variables that could possibly help develop accurate predictive models of ratings in the later stages of the project.

4. Hypothesis Testing

4.1 Correlation Analysis

Hypothesis Testing was performed to find out if there are statistically significant relationships among the selected variables and the IMDb rating variable. Pearson correlation analysis was used to determine the degree of association between numeric variables such as number of votes; movie run time in minutes; start year of the movies, and ratings.

For the relationship between votes and ratings, the hypotheses were defined as follows:

H0: There is no relationship between the number of votes and ratings.

H1: There is a relationship between the number of votes and ratings.

Pearson correlation testing resulted in a p value below 0.05. This means the Null Hypothesis can be rejected; therefore, it appears there is a statistically significant relationship between the number of votes and the movie ratings. However, the correlation coefficient indicated that the relationship is relatively weak.

In addition to analyzing votes, similar analyses were done on runtime and start year. The results indicate that neither runtime nor start year has a particularly strong linear association with ratings even though there weak correlations may still exist.

4.2 ANOVA Test for Genres

A one-way ANOVA test was performed to examine whether average ratings differ significantly across genres.

The hypotheses for the ANOVA test were defined as follows:

H0: There is no difference in average ratings across genres.

H1: There is a difference in average ratings across genres.

The ANOVA results produced a p-value greater than 0.05. Therefore, the null hypothesis is failed to be rejected. This shows that there is no statistically significant difference in average ratings between the tested genres based on the data.

However, this result does not necessarily mean that genre information is not important for prediction tasks. Genre variables may still contribute to machine learning models when properly encoded and combined with other variables.

4.3 Interpretation of Statistical Results

Overall the Hypothesis Testing process showed that there is statistical evidence for some of the variables having a relationship with the ratings but there was no evidence to support any one variable being dominant for rating a TV Series. The results indicate that audience evaluation behavior can be explained using multiple factors; therefore machine learning models may provide better predictive results and capture more complex relationships within the data.

5. Machine Learning Models

5.1 Data Preparation

Prior to developing the machine learning models, the data set was prepared for use in predictive modeling. The numerical variables on this data set such as votes, runtime, startYear, and ratings had to be appropriately formatted into numbers.

The missing values in key variables were deleted so that the model could train without using any incomplete records.

Genre information also needed some form of preprocessing. Genre is an example of a categorical type of variable where one TV program can have many different genres. Thus, the genre information was transformed into binary format via encoding so it would be able to be inputted along with other numerical variables into the machine learning models.

Once all of the necessary transformations were completed, the data set was split into two separate groups; a test group and a training group. For our purposes, approximately 80% of the data was allocated to the training group and the remaining 20% to the test group.

5.2 Linear Regression Model

Linear Regression was used as a baseline machine learning model to predict IMDb ratings. The model attempts to learn how well the predictor variables can be related to the target variable through a linear relationship.

Votes, runtime, release year and encoded genre variables were the features used in this model. After the model had been trained, predictions were generated on the test data set. Performance of these predictions were then assessed with performance metrics R^2 , MAE and MSE.

Performance of the Linear Regression model was moderate. This indicates that simple linear relationships alone are not sufficient to fully explain TV series ratings.

5.3 Random Forest Model

In addition to Linear Regression, a Random Forest Regressor model was applied to enhance the quality of predictive performance. Random forest is a different type of machine learning model than linear regression. It has more flexibility to recognize or identify relationships that are non linear or have many interactions between variables.

Both models used the exact predictor variables. Random forest yielded a better performance with respect to prediction when compared to linear regression. R^2 scores improved and error rates slightly decreased. These results suggest the data contains complex patterns that were not completely recognizable using a simple linear relationship.

5.4 Model Evaluation

Multiple performance metrics were used for the evaluation of the machine learning models. The R^2 value is a measure of how much variance in rating can be attributed to the models. In addition to measuring variance, Mean Absolute Error (MAE), and Mean Squared Error (MSE) were employed as measures of the prediction accuracy of the models.

In comparison with the linear regression model, the random forest model demonstrated greater R^2 values and lower error in its predictions, thereby demonstrating better performance overall. Despite the improvements in performance, the overall predictive capability of the models remains moderately good. Therefore, it is likely that some other factors influence ratings of TV series, in addition to those measured by the data.

5.5 Feature Importance Analysis

Feature importance analysis was performed using the Random Forest model to identify which variables contributed most to prediction performance.

The results show that the total number of votes a TV show has received is one of the top contributing factors to how well a TV show can be predicted based upon its ratings.

Runtime and start year are other important variables when it comes to making predictions. Some genre categories like drama and reality TV appear in the list of important factors but tend to have less influence than the key numerical variables.

Overall, the feature importance analysis indicates that both numeric values and encoded information regarding genres affect how well TV shows will be rated.

6. Findings and Discussion

The analytical work done during this project produced many important insights into how well rated are the Turkish TV Series analyzed as well as what variables contribute to audience perceptions and evaluations of them.

Firstly; an initial exploration of the data indicated that ratings for most television programs fall within moderate to high rating values. However, there is considerable variation in the number of votes on TV Series. This implies that there may be large variations in how engaged audiences are with different television programs.

Correlation analysis among the data revealed a weak positive relationship between the vote count for each show and its respective rating from IMDb. While the statistical significance of this relationship was established, the magnitude of the correlation coefficient was low. In essence, being popular does not appear to be enough to completely account for how viewers rate a TV show.

Also, the data analysis revealed that neither runtime nor release year had strong relationships with ratings. Also, an ANOVA test failed to find any statistically significant differences in mean ratings between genres.

However, the analyses using machine learning found that even though the genre information added to prediction performance when it was properly encoded and combined with other variable, the addition of the encoded genre variables did add some value to both the richness of the feature set and to a slight improvement in overall model performance.

Of all the machine learning algorithms used, Random Forest performed better than Linear Regression. Therefore, it appears that there are non-linear patterns and more complex relationships between variables and ratings present in the dataset.

In analyzing the feature importance of the various models used, it was evident that the number of votes were a key predictor in nearly all models. However, runtime, starting year, and several genre categories were also contributing factors to the prediction process but they contributed less than vote count.

Therefore, collectively these findings support the conclusion that ratings of TV Series are determined by multiple interconnected factors. None of the individual factors evaluated here could independently predict how well a TV Series would be perceived and evaluated by an audience.

7. Limitations and Future Work

Although this project provides us interesting insight for how viewers rate Turkish TV Series, there are some significant limitations of this project that needs to be considered.

A large limitation is the sample size and scope of data used. The project only analyzed a few variables that may be related to ratings; however it does not include all possible variables that affect ratings.

Additionally, the data set did not have information on other factors, which might greatly affect ratings, such as actors involved in making the show, director, budget for producing each episode, number of episodes produced, popularity of the streaming service where the show is located, demographic characteristics of the viewing audience, and viewers use of social media to discuss the show. These can greatly impact what viewers think about and ultimately view.

Also, although the model was able to predict moderately well, there is still room for improvement. It appears that audiences rating behaviors towards TV shows are complicated and influenced by many unknown or subjectively determined elements that cannot easily be measured.

Future studies could address these limitations through larger samples of data, additional relevant data points or more sophisticated machine learning algorithms. Techniques like deep learning algorithms, recommendations systems and/or natural language processing algorithms could be employed to analyze reviews or comments made by viewers regarding their experience watching TV Series.

Furthermore, future research could compare Turkish TV series with international productions in order to better understand differences in audience preferences and rating behavior across cultures.

8. Conclusion

This research on the factors influencing the popularity of Turkey's TV Series examined this topic through data analysis, hypothesis testing, and machine learning.

Variables like vote count, runtime, release year, and genre were found to have an influence on predicting IMDB ratings, but many of these relationships remained weak. The analysis also indicated that there is no one factor for determining TV Series', there are in fact many factors combined that does.

Of all of the machine learning algorithms used in the study, Random Forest was found to perform at a higher level than linear regression, indicating some non-linearity exists within the data set.

In conclusion, this project demonstrated how Data Science tools can be applied to datasets of entertainment related content to examine audience behavior and prediction. While the dataset has its own limits, the project did demonstrate both how statistical methods can be used alongside machine learning methodologies to provide insight into Turkish TV Series' ratings.